

HUAILAI, China

WMO: 544050

Lat: 40.4174N

Lon: 115.5071E

Elev: 571

StdP: 94.65

Time Zone: 8.00 (E08)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -15.1	(c) -13.4	(d) -30.3	(e) 0.2	(f) -9.7	(g) -27.8	(h) 0.3	(i) -8.6	(j) 12.0	(k) -10.7	(l) 11.0	(m) -9.4	(n) 4.8	(o) 290	(p) 0.447

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 7	(b) 9.5	(c) 33.5	(d) 19.5	(e) 32.0	(f) 19.6	(g) 30.7	(h) 19.6	(i) 24.0	(j) 29.3	(k) 23.1	(l) 28.1	(m) 22.3	(n) 27.1	(o) 3.2	(p) 110

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
22.4	18.4	26.8	21.6	17.5	26.3	20.7	16.5	25.6	75.8	29.6	72.3	28.1	68.8	27.0	27.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%		2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
Min		Max		Min	Max		Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)		(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
9.9	8.5	7.4		-17.9	36.4	2.2	1.8	-19.5	37.6	-20.8	38.7	-22.0	39.6	-23.6	40.9	
				-18.9	25.3	2.1	0.9	-20.4	25.9	-21.6	26.5	-22.8	27.0	-24.3	27.6	
			DB													
			WB													

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	10.7	-6.3	-2.5	4.8	12.7	18.9	23.1	25.2	23.7	18.6	11.2	2.4	-4.5
	DBStd	11.62	3.69	4.01	4.54	4.34	3.55	2.86	2.30	2.24	3.18	3.73	4.37	3.63
	HDD10.0	1757	506	350	172	22	1	0	0	0	30	228	449	
	HDD18.3	3421	764	583	421	173	35	2	0	1	34	222	477	708
	CDD10.0	2003	0	0	9	103	277	393	470	424	258	67	1	0
	CDD18.3	626	0	0	0	5	53	146	212	166	42	1	0	0
	CDH23.3	5565	0	0	3	85	568	1336	1941	1297	324	12	0	0
	CDH26.7	1943	0	0	0	17	183	516	749	406	70	1	0	0
Wind	WSAvg	2.8	3.6	3.3	3.3	3.2	3.1	2.4	2.2	2.1	2.1	2.5	2.8	3.4
Precipitation	PrecAvg	387	2	3	7	16	26	55	112	101	39	17	4	1
	PrecMax	544	13	25	22	76	62	134	263	200	86	50	19	5
	PrecMin	192	0	0	0	0	2	12	43	21	5	0	0	0
	PrecStd	83	3	5	6	15	14	29	56	48	22	14	4	1
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	6.1	11.2	21.8	28.7	33.5	35.5	35.9	33.7	31.3	24.7	16.4	7.0
		MCWB	-0.5	2.7	9.2	13.5	16.0	17.6	20.6	21.4	18.1	14.1	7.1	0.4
	2%	DB	3.3	8.7	17.9	25.6	30.8	33.2	33.7	31.7	28.5	22.0	13.6	4.6
		MCWB	-2.1	1.4	6.9	11.8	15.1	18.1	20.9	21.1	17.7	12.3	5.8	-0.9
	5%	DB	1.4	6.5	15.2	23.2	28.7	31.5	32.0	30.3	26.7	20.0	11.1	2.8
		MCWB	-3.3	0.2	5.8	11.0	14.5	17.9	21.1	21.0	16.9	11.3	4.6	-2.1
	10%	DB	-0.3	4.5	12.6	20.8	26.6	29.7	30.6	29.0	25.1	17.9	9.1	1.2
		MCWB	-4.5	-0.9	4.3	9.9	13.8	17.8	21.2	20.6	16.2	10.6	3.7	-3.1
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	-0.2	3.9	10.2	16.0	19.1	22.3	25.2	25.1	21.6	16.1	8.4	0.9
		MCDB	5.3	8.7	18.7	22.7	27.1	29.1	30.6	30.2	27.2	21.3	13.0	5.7
	2%	WB	-1.9	1.9	8.0	14.0	17.7	21.3	24.2	23.9	20.0	14.5	7.0	-0.7
		MCDB	2.9	8.0	16.7	22.2	25.9	27.9	29.8	28.7	24.8	19.3	11.7	4.0
	5%	WB	-3.2	0.5	6.2	12.5	16.6	20.4	23.5	23.0	18.8	13.1	5.5	-1.8
		MCDB	1.1	6.1	14.1	20.6	24.2	26.8	28.9	27.5	24.0	17.6	10.4	2.3
	10%	WB	-4.4	-0.8	4.8	11.1	15.5	19.6	22.8	22.1	17.7	11.7	4.0	-2.9
		MCDB	-0.5	4.2	11.9	18.8	23.2	25.9	27.8	26.4	23.0	16.2	8.7	0.9
Mean Daily Temperature Range	5% DB	MDBR	8.9	10.2	11.2	12.0	12.2	11.1	9.5	9.9	11.0	10.6	9.1	8.3
		MCDBR	10.8	12.9	14.8	15.9	16.1	14.5	12.6	12.0	13.5	13.4	11.9	10.2
	5% WB	MCWBR	7.4	8.0	7.9	6.9	5.9	4.4	3.8	4.1	5.0	6.1	6.7	6.7
		MCDWR	10.6	12.3	13.7	13.3	12.3	10.6	9.3	9.2	10.5	11.1	10.9	10.0
Clear-Sky Solar Irradiance	taub		0.351	0.438	0.464	0.535	0.592	0.714	0.690	0.668	0.579	0.506	0.391	0.359
	taud		2.139	1.931	1.909	1.809	1.708	1.540	1.624	1.645	1.766	1.862	2.103	2.162
	Ebn at Noon		775	742	775	749	717	632	642	639	664	661	717	730
	Edh at Noon		107	154	175	207	234	277	253	241	200	163	111	97
All-Sky Solar Radiation	RadAvg		2.63	3.48	4.74	5.63	6.23	5.64	5.28	5.12	4.46	3.55	2.67	2.31
	RadStd		0.14	0.23	0.36	0.35	0.45	0.48	0.32	0.31	0.24	0.27	0.29	0.15

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (7 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page